Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/26/2006
PI Name:	Bunick, Gerard J. Ph.D.		
Project Title:	Structural Studies of Nucleosomes and Chromatin		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Oak Ridge	**State:**	TN
Zip Code:	37830-8026	**Congressional District:**	3
Comments:	Deceased September 2007.		
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	10/01/2001	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Miller, Teresa	**Contact Phone:**	256-544-7815
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1826		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This research program is exploring the production of large nucleosome core particle crystals suitable for neutron diffraction to examine the role of hydration in chromatin, and to incorporate H3-replacing centromeric proteins in the histone octamer and reconstituted nucleosome core particles (NCP). Neutron structures of D-xylose isomerase are being determined and analyzed to reveal the protonation states of active-site amino acid residues and water molecules. The first task is to conduct research to incorporate H3-replacing centromeric proteins in the histone octamer and reconstituted nucleosome core particles (NCP). The structures will provide additional critical insights into structure-function relationships in chromatin. The second task will elucidate the mechanism of D-xylose isomerase using information obtained from neutron diffraction studies. Structures of modified nucleosome core particles are important to the elucidation of the molecular processes of gene regulation and transcription in chromatin, the substrate for all such activity in humans. The knowledge acquired from this research may provide answers to the genesis and proliferation of cancer and other diseases. The studies of D-xylose isomerase demonstrate the elucidation of enzyme mechanisms by		

	combining neutron diffraction with traditional X-ray structure determination.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Structural studies are demonstrating the power of combining X-ray diffraction results with spallation neutron diffraction for the elucidation of enzyme mechanisms. Our neutron results may encourage the construction of new macromolecular diffraction instruments at spallation sources now under construction in the United States and abroad. The proposed research will extend our knowledge of chromatin structural biology, providing information linking the structure of chromatin to the regulation of gene expression and control. The results may help to understand aspects of cancer and other diseases.
Task Progress:	The H3 like centromeric histones, DH6, YMH3, chicken CenpA, and human CenpA are the target of expression and incorporation into histone octamer, replacing the normal component, histone H3. A clone for the non-histones high mobility group protein HMGN1 was prepared expression demonstrated. The HMGN1 will be complexed with nucleosome core particles. A neutron structure of D-xylose isomerase has been completed (PDB code:2GVE). A 0.94 Angstroms resolution X-ray structure of the enzyme at 100K has been completed (PDB code:2GLK). The neutron studies were among the first to use spallation-produced neutrons and time-of-flight diffraction measurement on such a large macromolecule. The research has been published in Proceedings of the National Academy of Sciences. Neutron and X-ray structures of the enzyme with various ligands will be determined in the future. Beam time at the Los Alamos Neutron Science Center (LANSCE) Protein Crystallography Station (PCS) is scheduled in 2006.
Bibliography Type:	Description: (Last Updated: 11/16/2007)
Abstracts for Journals and Proceedings	Glusker JP, Katz A, Carrell HL, Li X, Hanson B, Langan P, Schoenborn BP, Bunick GJ. "Time-of-flight neutron diffraction as an aid to elucidating enzyme mechanisms: D-xylose isomerase." Conference on New Frontiers in Neutron Macromolecular Crystallography, July 12-13 2005. Conference on New Frontiers in Neutron Macromolecular Crystallography, July 2005. http://www.sns.gov/workshops/mandi2005/index.shtml#talks , Jul-2005
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Abstracts for Journals and Proceedings	Bunick GJ, Katz A, Li X, Carrell HL, Hanson B, Langan P, Coates L, Schoenborn BP, Glusker JP. "Locating Acitive-site Hydrogen Atoms in D-Xylose Isomerase: Time-of-Flight Neutron Diffraction." American Conference on Neutron Scattering, St. Charles Illinois, June 18-22, 2006. Session: Protein Structures, Mechanisms and Functional Complexes. American Conference on Neutron Scattering, June 2006. http://acns2006.anl.gov/ , Jun-2006
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Articles in Peer-reviewed Journals	Young GW, Hoofring SA, Mamula MJ, Doyle HA, Bunick GJ, Hu Y, Aswad DW. "Protein L-isoaspartyl methyltransferase catalyzes in vivo racemization of Aspartate-25 in mammalian histone H2B." J Biol Chem. 2005 Jul 15;280(28):26094-8. PMID: 15908425 , Jul-2005
Articles in Peer-reviewed Journals	Katz AK, Li X, Carrell HL, Hanson BL, Langan P, Coates L, Schoenborn BP, Glusker JP, Bunick GJ. "Locating active-site hydrogen atoms in D-xylose isomerase: Time-of-flight neutron diffraction." Proc Natl Acad Sci U S A. 2006 May 30;103(22):8342-7. PMID: 16707576 , May-2006
Articles in Peer-reviewed Journals	Bunick GJ, Hanson BL. "Annealing macromolecular crystals." Methods in Molecular Biology/ Macromolecular Crystallography Protocols, Volume 2, 2006 Oct, p. 1-16. , Oct-2006